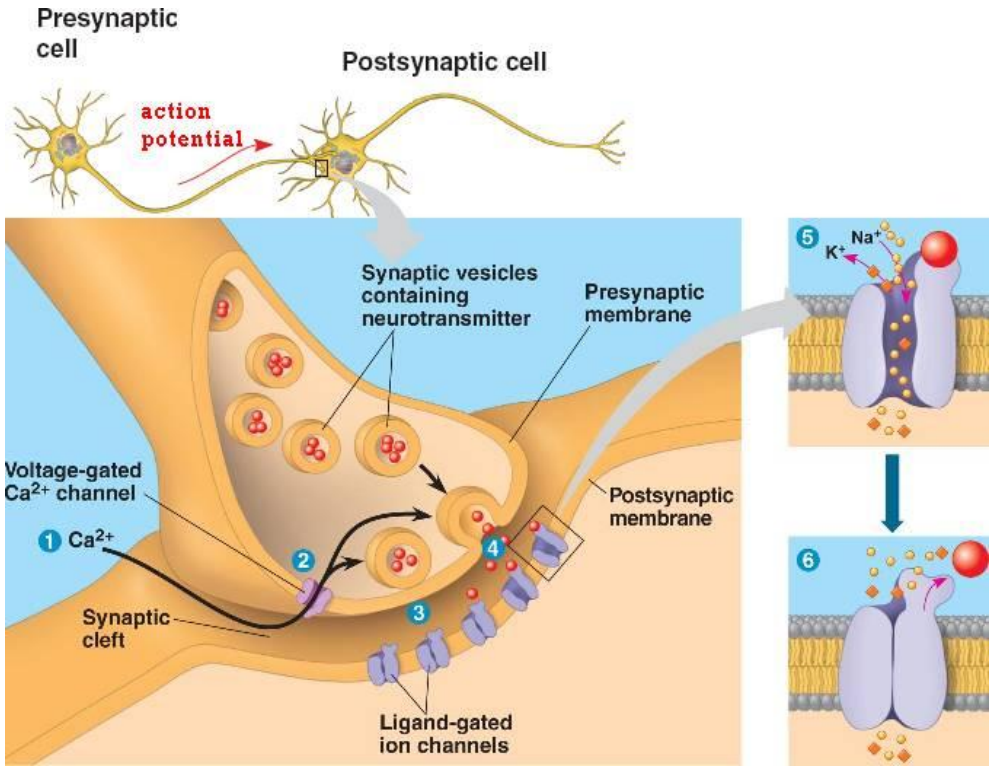
# Ion channel-mediated signaling

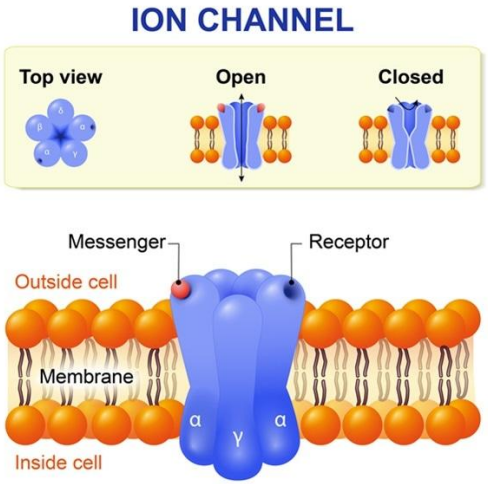



**OVERVIEW**

- Relevance and characteristics of ion-channel signaling.
- Acetylcholine (Ach) receptor at the neuromuscular junction.
- Alterations of the Ach signaling.
- Botox.

# Relevance and main characteristics

**-Relevance:** *Present in all cells. Particular importance in <u>electrically excitable cells</u> (neurons, muscle and some endocrine cells)*




**-Characteristics:**

*-Transmembrane proteins*

*-Down the concentration gradient*

*-Very high rate of transport (>$10^6$ ions/sec)*

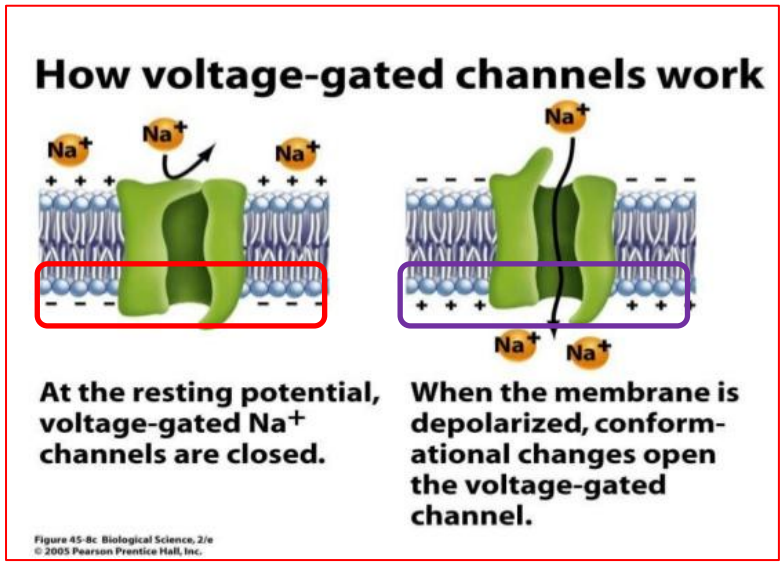
*-Two major types:*

*a) **Voltage-gated:** The channel is regulated by changes in the membrane potential*

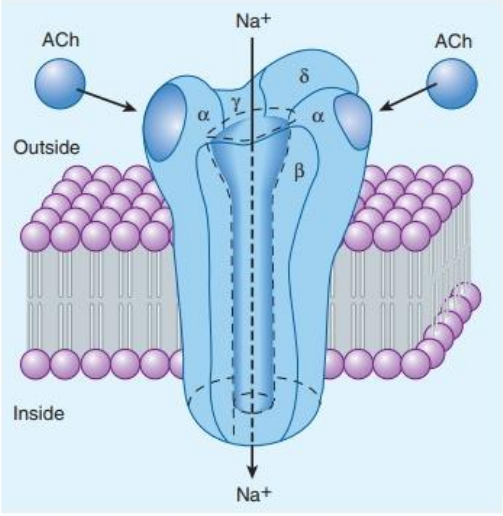
*b) **Ligand-gated:** The channel is regulated by binding of a ligand*

*-Main ions involved: $Na^+$, $K^+$, $Cl^-$, $Ca^{+2}$*

-Voltage-gated ion channels.

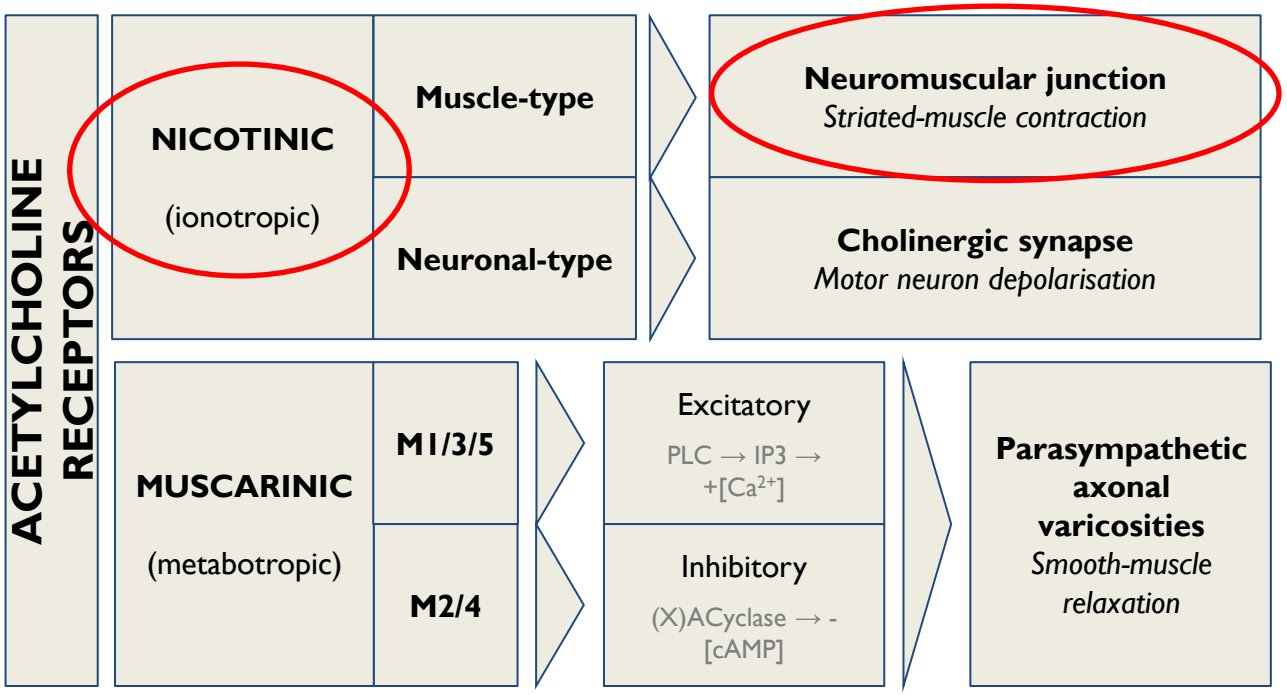



-Ligand-gated ion channels. Focus on Na+ channels regulated by Acetylcholine (Ach)



# Acetylcholine receptors

Do not study this




**Plasticity of Ach signaling: activity through different receptors**

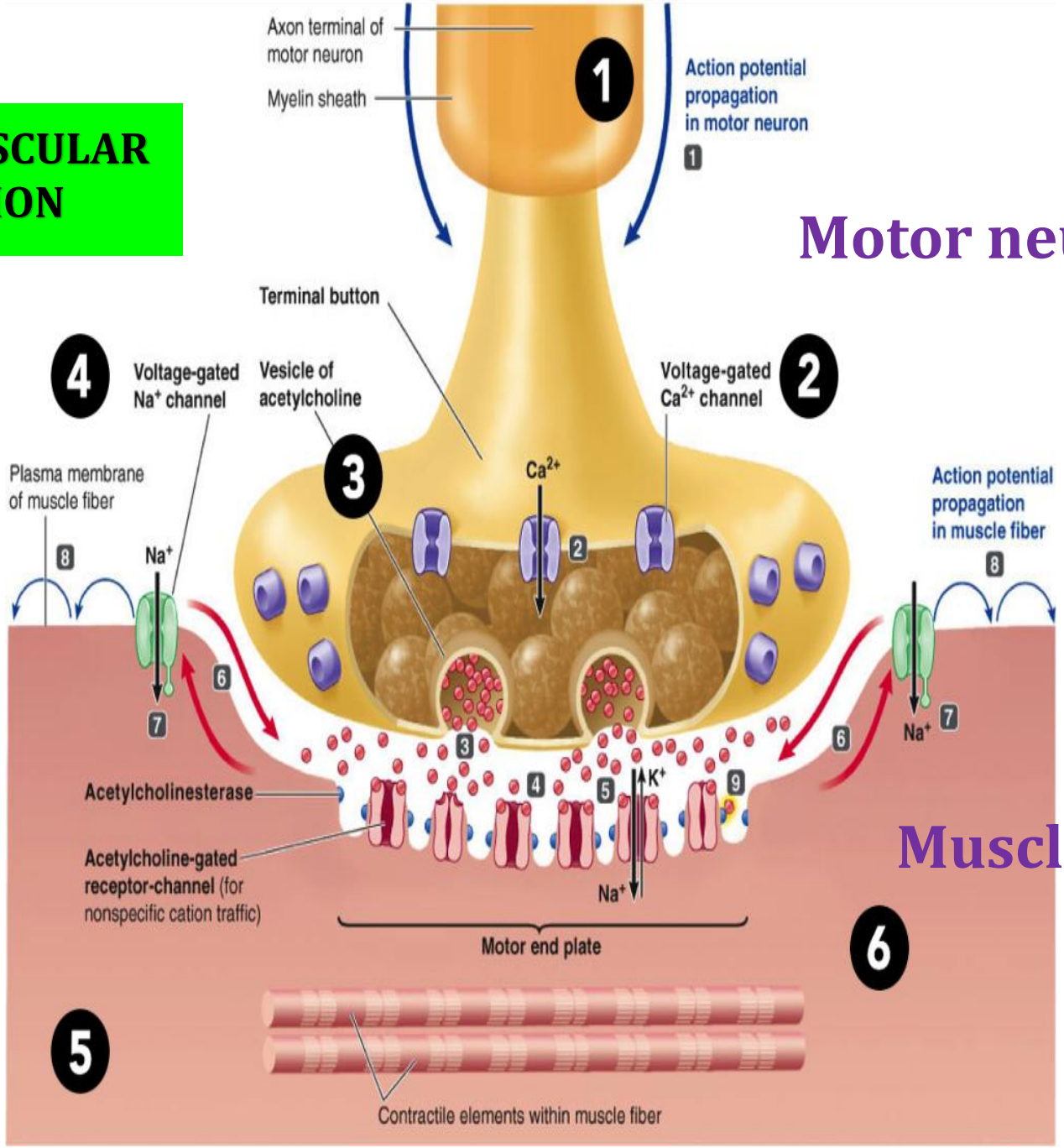
-Nicotinic: **Ach** binds to an ion channel receptor. It is also activated by nicotine.

-Muscarinic: **Ach** binds to a G Protein Coupled receptor (GPCR). It is also activated by muscarine.



*Amanita muscaria*

# NEUROMUSCULAR JUNCTION

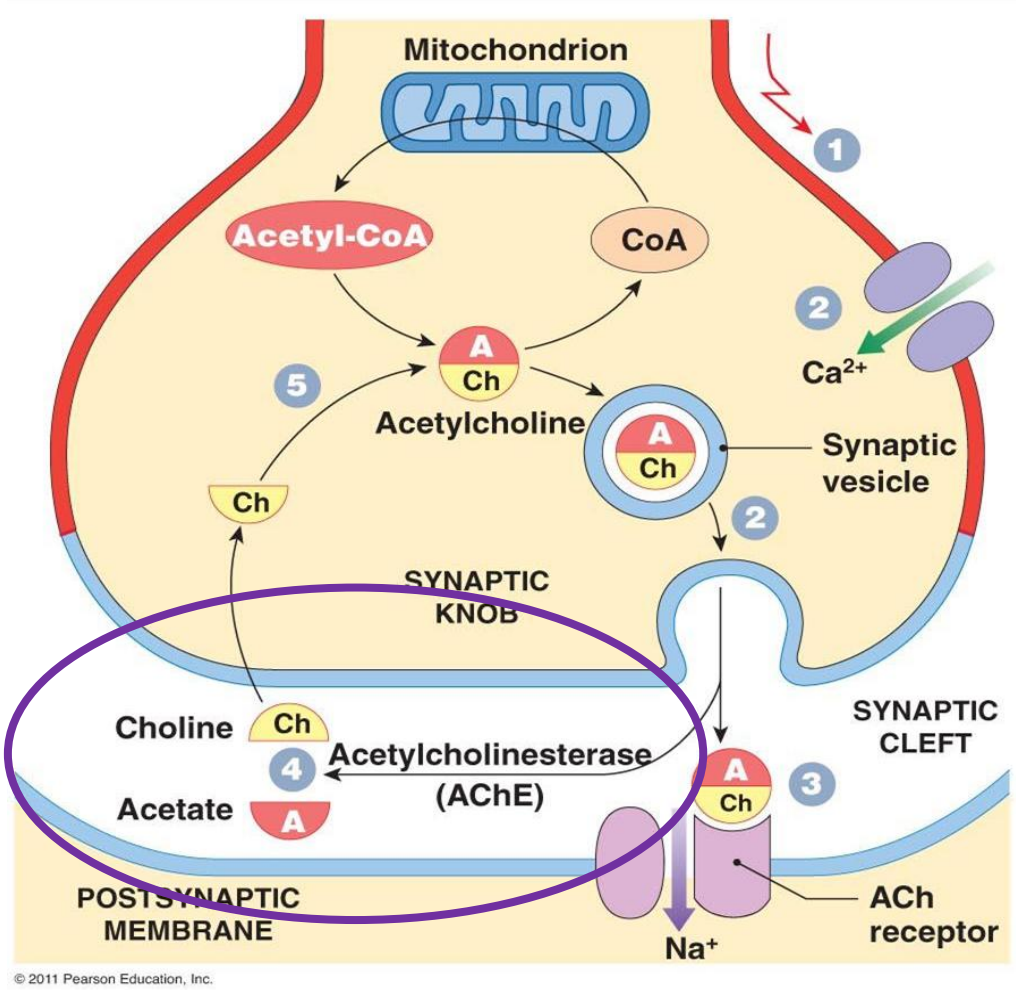


# Video on motor neuron/ muscle fiber signaling

https://www.youtube.com/watch?v=eTYe1CtjJRE



# Acetylcholinesterase: stopping signaling

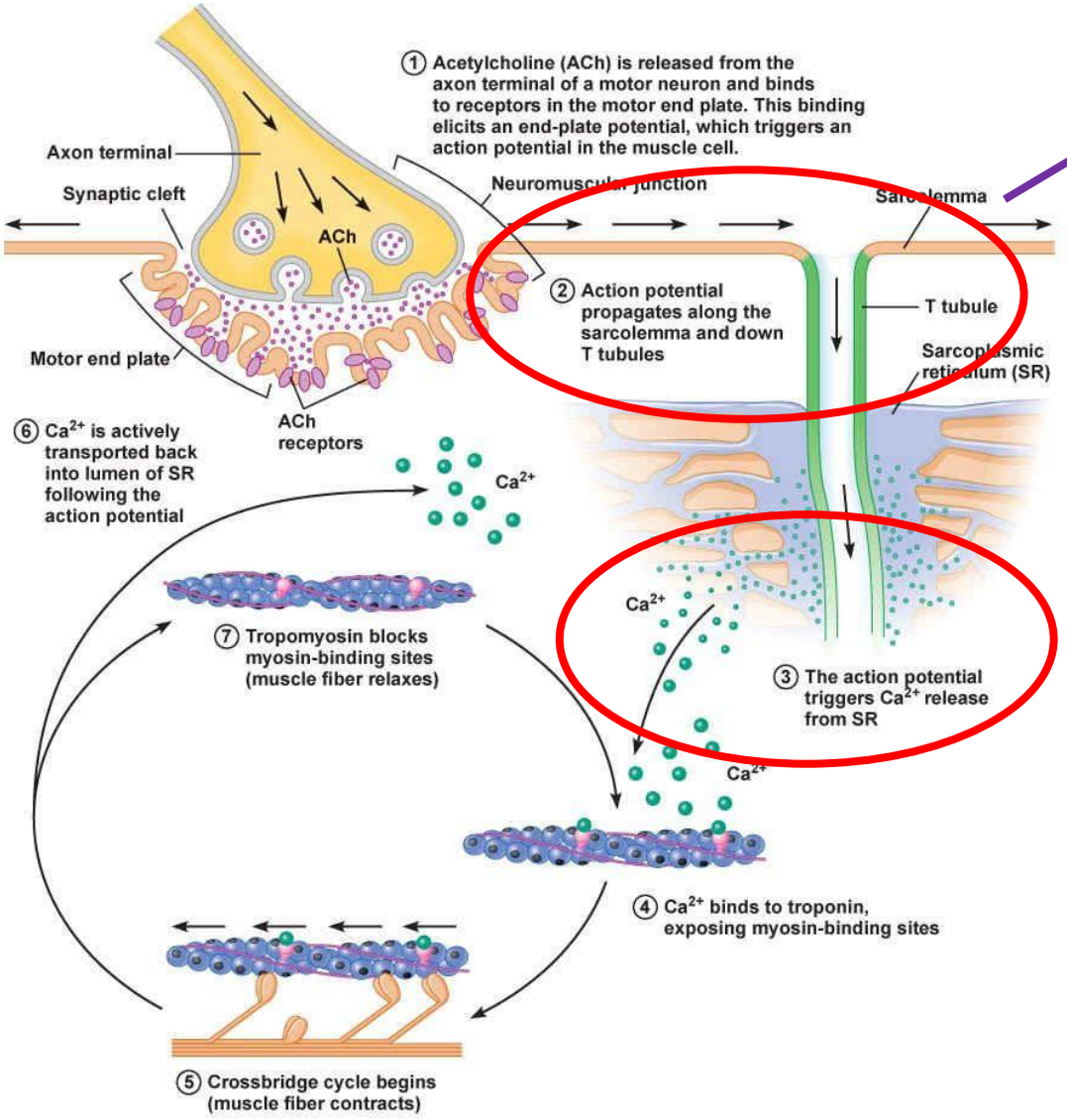



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**Events Occurring at Synapse**

1. An arriving action potential depolarizes the synaptic knob.
2. Calcium ions enter the cytoplasm, and after a brief delay, ACh is released through the exocytosis of synaptic vesicles.
3. ACh binds to sodium channel receptors on the postsynaptic membrane, producing a graded depolarization.
4. Depolarization ends as ACh is broken down into acetate and choline by AChE.
5. The synaptic knob reabsorbs choline from the synaptic cleft and uses it to synthesize new molecules of ACh.

# Summary of events at the fiber: muscle contraction



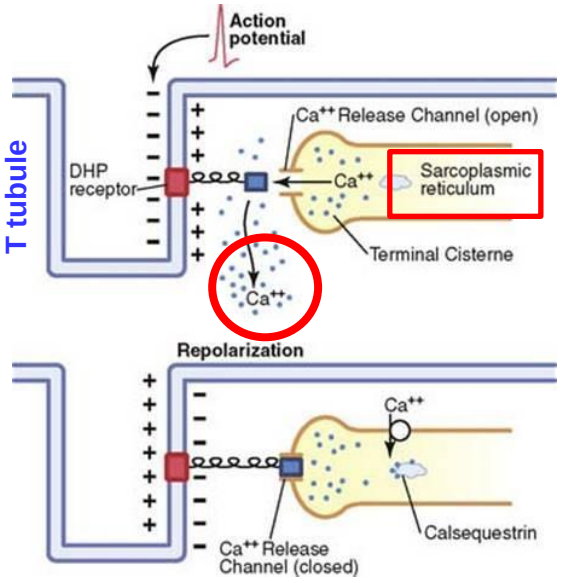

Action potential propagates through the membrane invaginations (**T tubules**)

↓

**T tubules**, in contact with the **sarcoplasmic reticulum (SER): Ca+2 channels open**

↓

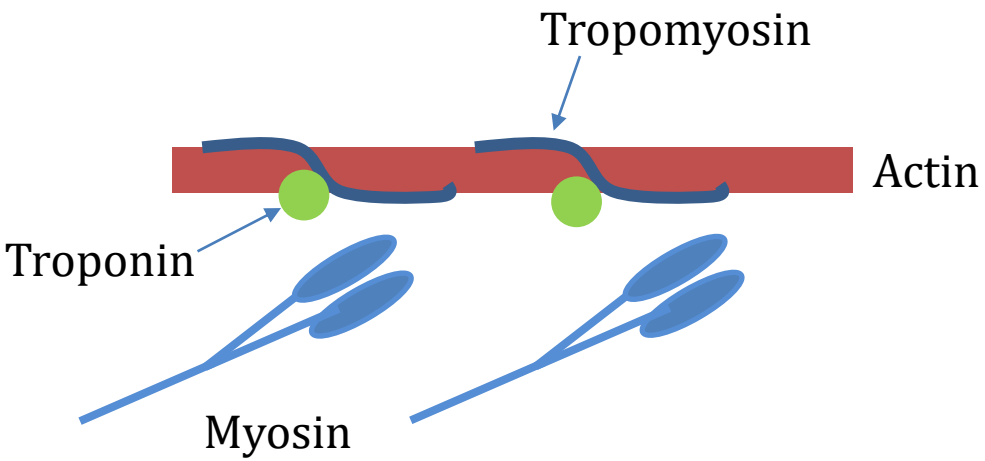
**Increase in cytosolic Ca+2**



# In the Sarcomere

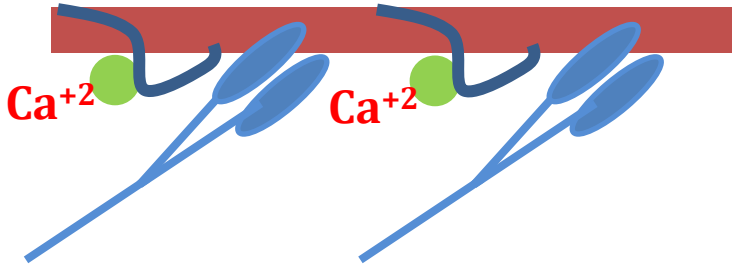
## Muscle relaxed

**LOW Ca+2**




## Muscle contracted

**HIGH $Ca^{+2}$**

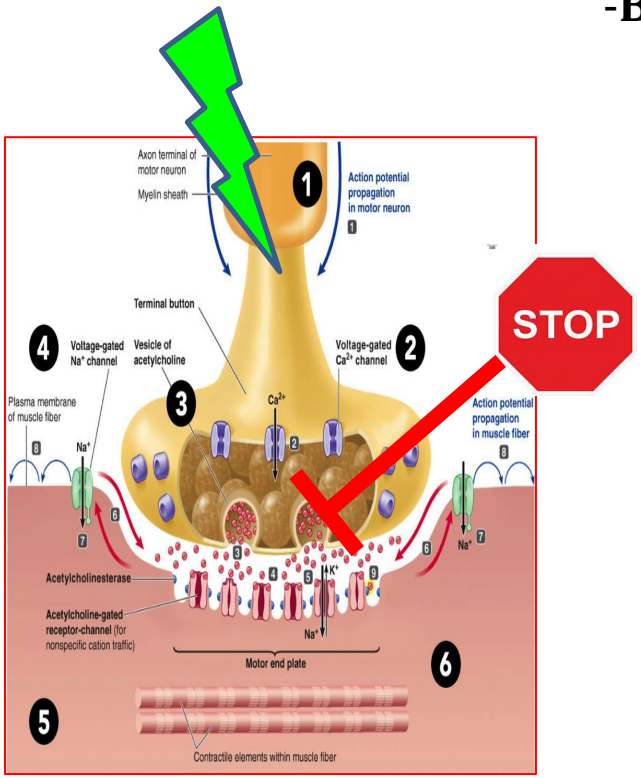


# Some toxins that alter Ach signaling at the neuromuscular junction

**-Tetanus toxin**
(*Chlostridium tetani*): **TETANUS**



**Causes hyperactivation of the motor neuron: Too much Ach release**




**-Botulinic toxin (Botox)**
(*Chlostridium botulinum*):
**BOTULISM**

**Blocks exocytosis of Ach-containing vesicles**



Rigidity, muscle spams, paralysis
https://www.youtube.com/watch?v=b2JLTrfGOoo



Muscle inactivity, many times fatal

# Botox: from toxin to treatment

**Cosmetics**




https://www.youtube.com/watch?v=gUE6NhltGbw

**-Muscle spasms**

**-Migraines**

**-Hyperhidrosis**





# Interfering with Ach signaling: poisoning and chemical weapons…

**Assassination of Kim Jong-nam, brother of Kim Jong-un, in 2017**








Probably killed with VX (nerve agent): an acetylcholinesterase inhibitor



Cobra venom: a nicotinic acetylcholine receptor antagonist

**Honey badger vs. cobra venom**